

Intelligent Question-Answering on Geomorphology Knowledge Based on Knowledge Graph Retrieval-Augmented Generation Technology

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Abstract

Geomorphology is a discipline dedicated to the study of the morphological characteristics, genesis, distribution, and evolutionary patterns of the Earth's surface. Its intrinsic disciplinary attributes and research domains underscore its pivotal foundational role within the field of geography. However, general-purpose large language models lack training in vertical domains, resulting in suboptimal performance in the field of geomorphology-related question answering. By leveraging knowledge graph retrieval-augmented generation technology, we construct knowledge graphs and knowledge graph communities. Through the integration of graph structures and external knowledge bases, we enhance the responses of large language models, achieving a deep fusion between large language models and knowledge graphs. Using karst landform knowledge as an experimental case for validation, we employed a comparative analysis approach to evaluate the question-answering performance of large language models based on knowledge graphs from both subjective and objective dimensions. The results indicate that, compared to traditional retrieval-augmented generation, knowledge graph retrieval-augmented generation technology demonstrates improvements in the logicity, depth of knowledge, and interpretability of responses. This advancement provides a novel tool for knowledge discovery and reasoning in geomorphological research.

1 Introduction

The study of geomorphology is both a basic and applied science that is devoted to understanding Earth surface processes, system response, and landscape evolution. Geomorphology may also be linked to the needs of society (Keller et al, 2020). The disciplinary attributes and scope of geomorphology underscore its critical role in geographical science research. Nonetheless, challenges such as inconsistent descriptive standards and semantic heterogeneity complicate the association, integration, analysis, and sharing and reuse of geoscience data (Xiong et al, 2021). Addressing these challenges necessitates the development of a unified, formalized, and machine-interpretable semantic framework for geographical research in the big data era, leveraging knowledge engineering techniques (Zhou et al., 2021).

The geomorphology knowledge graph represents an extension of knowledge graph methodologies into the domain of geographical research. It serves as a formalized system for describing geomorphological knowledge, entities, and their relationships (Chen et al., 2017), offering systematic and profound insights into geomorphological phenomena (Ballatore et al., 2015). This approach demonstrates significant potential across various applications, including geomorphological knowledge analysis, geoscience problem-solving, spatiotemporal forecasting, and decision-making. It plays a pivotal role in advancing geography from traditional information services towards advanced knowledge services (Lu et al., 2017).

Regarding geomorphological knowledge integration, methods typically draw upon generalized knowledge alignment systems, distinguishing between data-layer and conceptual-layer integration. Data-layer integration focuses on aligning instances, employing techniques such as entity linking and resolution to unify specific instances (Jin et al., 2019). Conceptual-layer integration involves aligning entities and integrating knowledge across languages (Zhuang et al., 2016). Although knowledge graphs excel at problem-solving and extracting answers, they face challenges related to the variability and ambiguity of natural language. Particularly, capturing the deeper semantics within complex queries and enhancing the efficiency of question-answering during intricate reasoning processes remain significant research hurdles that need addressing.

Currently, large language model (LLM) technology has achieved significant advancements in natural language processing. In October 2022, OpenAI introduced ChatGPT, heralding a new era for question-answering systems. Trained on vast quantities of internet data, these models can now boast parameters numbering in the trillions, with some reaching tens of trillions, significantly enhancing their generative capabilities. Since then, numerous models have been developed and deployed globally, including Flan T5 (Chung et al., 2022), PaLM (Chowdhery et al., 2023), the Llama series, the GPT series, Baidu's Wenxin Yiyang (Sun et al., 2021), iFlytek's Xinghuo cognitive model, Huawei's Pangu model, the ChatGLM series, BaiChuan, among others. These LLMs have demonstrated effectiveness across various downstream tasks, particularly excelling in text classification, named entity recognition, text summarization, and question answering.

As research has progressed, it has become evident that large language models (LLMs) struggle with issues such as poor explainability, inadequate robustness, and limited reasoning capabilities. When confronted with queries that they have not encountered during training or that extend beyond the scope of their training data, these models may generate responses that appear plausible but are, in fact, inaccurate or entirely fictitious. This tendency arises from their formidable language generation capabilities and their capacity to generalize over language patterns, resulting in what is known as the "hallucination" phenomenon (Ji et al., 2023).

To address the "hallucination" issue, approaches can be categorized into two methodologies: domain-specific fine-tuning and retrieval-augmented generation (RAG). Domain-specific fine-tuning involves training existing language models on additional data to optimize them for particular tasks. Unlike comprehensive fine-tuning of large language models (LLMs), which demands vast amounts of high-quality data and significant computational resources—and often results in catastrophic forgetting during the process—efficient domain-specific fine-tuning reduces the number of parameters needing adjustment and lowers computational complexity. This approach enhances LLM performance on targeted tasks while mitigating the risk of catastrophic forgetting. Popular strategies for efficient domain-specific fine-tuning include Adapter, Prefix Tuning, and LoRA techniques. Notable examples encompass AdapterP (Pfeiffer et al., 2020), Parallel (He et al., 2021), P-Tuning (Liu et al., 2023), AdaLoRA (Zhang et al., 2023), and QLoRA (Dettmers et al., 2024). The essence of these domain-specific fine-tuning efforts lies in leveraging LLMs by incorporating specialized domain data to boost performance within that domain. However, given the highly intricate architecture of LLMs, challenges such as overfitting to a specific domain or task and forgetting general knowledge acquired during pre-training can arise during fine-tuning.

RAG represents a concrete realization of the concept of infusing domain-specific knowledge contexts. It employs information retrieval techniques to extract information segments that are intimately connected to the core of the query from an external knowledge repository, which are then fed as reference data into large language models to bolster the precision and dependability of the responses (Fan et al. 2024; Gao et al. 2024; Hu et al. 2024). At the heart of retrieval-augmented generation technology lie text embedding models and semantic retrieval methodologies. Noteworthy models encompass Word2vec (Mikolov et al., 2013a), GloVe (Pennington et al., 2014), BERT (Devlin et al., 2018), and E5 (Wang et al., 2022). Semantic search technology capitalizes on sophisticated natural language processing capabilities to profoundly comprehend the underlying semantic essence of query texts and potential documents, thereby facilitating an accurate alignment of user query intentions with the genuine semantics of documents (Mikolov et al., 2013b). Exemplary techniques include Maximum Inner Product Search (Ram et al., 2012) and Approximate Nearest

Neighbor Search (Malkov et al., 2018). These technologies empower models to dynamically tap into and exploit external knowledge repositories to acquire more extensive and pertinent knowledge, consequently producing responses that are more aligned with factual accuracy. However, RAG technology underperforms in addressing Query-Focused Summarization (QFS) tasks (Nelson, 2024). To address this limitation, Graph-based Retrieval-Augmented Generation (Graph RAG) has been proposed. This approach integrates knowledge graphs, retrieval-augmented generation, and query-focused summarization, producing responses that exhibit superior comprehensiveness and diversity (Edge et al., 2024).

To address the challenges in organizing and expressing geomorphological knowledge and enhancing machine comprehension of such knowledge, this study explores knowledge graph retrieval-augmented generation (GraphRAG) technology, focusing on the integration of knowledge graphs with large language models for intelligent question-answering. The research contrasts the response quality of traditional Retrieval-Augmented Generation (RAG) with that of GraphRAG. Geomorphological knowledge is inherently characterized by its spatiotemporal correlations, geographical specificity, multi-sourced nature, and high complexity. Traditional RAG methods show notable limitations in tasks requiring the synthesis of global datasets to produce accurate answers. Conversely, geomorphological knowledge graphs facilitate the effective integration of diverse and heterogeneous data through structured data integration mechanisms. Employing the GraphRAG framework to organize the knowledge base not only elucidates the temporal evolution of geomorphological entities but also enables detailed documentation of specific entities' contextual information, thereby improving the efficiency of understanding and analyzing complex geomorphological systems.

Secondly, the geomorphological knowledge graph not only systematically organizes geomorphological terminology and their definitions but also effectively illustrates the intrinsic characteristics and defining attributes of the knowledge these terms represent. While large language models inherently possess extensive knowledge, integrating them with geomorphological knowledge graphs to constrain their retrieval and reasoning boundaries enhances the relevance and accuracy of their outputs. This synergy ensures that the content generated by large language models remains grounded in reliable geomorphological facts.

More importantly, geomorphological knowledge is directly or indirectly linked to research questions in geomorphology. Whether explaining specific geomorphological phenomena or understanding the broader processes of geomorphological systems, such studies adhere to fundamental geomorphological principles and laws, forming a complex network of knowledge through interactions. This inherent logical interconnectedness necessitates systematic methods for integrating and reasoning with knowledge when addressing geomorphological problems. Consequently, the GraphRAG architecture plays a critical role, as it provides a high-level generalization and synthesis of geomorphological evolution processes, enabling systematic analysis and solutions to geomorphological problems across multiple scales and dimensions.

2 Methods

The geomorphological knowledge graph, as illustrated in Fig. 1, is a graph-structured knowledge base that enables efficient retrieval and intelligent reasoning. By leveraging the GraphRAG architecture, the integration of external structured geomorphological knowledge graphs enhances large language models' contextual understanding, thereby improving their ability to learn geomorphological knowledge. Embedding the geomorphological knowledge graph within the GraphRAG framework significantly enhances the

professionalism, credibility, authenticity, and interpretability of the responses generated by large language models in geomorphology.

2.1 Construction of Geomorphology Knowledge Graph

The construction and indexing of the geomorphology knowledge graph form the foundational basis of GraphRAG (as illustrated in Fig. 2), with the quality of the geomorphology knowledge graph exerting a direct influence on the performance of GraphRAG. To ensure high-quality recognition of geomorphological entities within documents by large language models and to enhance processing efficiency, two partitioning methodologies are utilized to logically segment the collected geomorphological literature into text chunks. These methodologies encompass sliding window partitioning and paragraph-based partitioning. Sliding window partitioning entails the establishment of a fixed window length that is subsequently slid across the text to facilitate segmentation, permitting overlapping sections between windows to preclude the omission of critical information within text chunks. Paragraph-based partitioning, on the other hand, segments the textual content by individual paragraphs, with each paragraph constituting a distinct text chunk. This approach is particularly apt for texts composed of small paragraphs, ensuring the preservation of the original semantic integrity of the text.

Each text chunk is linked to its corresponding document, and the text chunks are utilized to generate a vector representation of the document via average embedding, which facilitates the swift retrieval of implicit relationships between documents during the query phase. By harnessing the robust contextual comprehension and semantic analysis capabilities of large language models, graph extraction is conducted on the segmented text chunks, yielding a list of entities complete with names, types, and descriptions. A list of relationships, including sources, targets, and descriptions, is also extracted. Furthermore, large language models furnish concise summary descriptions for each entity and relationship. Given that identical geomorphological entities may be extracted from disparate text chunks, GraphRAG's merge operation is utilized to amalgamate multiple subgraphs into a new, more comprehensive graph. When identical nodes are encountered, a connection operation is executed to merge the corresponding attributes and relationships. For instance, in the extraction of the entity "Karst," as depicted in Fig. 3, post-merger, a list comprising multiple descriptions is generated: 'Karst terrain is categorized into Mediterranean, temperate, arid, monsoon, and tropical types, based on distinct karst terrain characteristics globally,' 'Karst landforms encompass stone buds, stalactites, blind valleys, and solution grooves,' 'Karst represents a distinctive geological and geographical phenomenon, whose inherent characteristics engender a range of unique phenomena impacting human economic activities.' The merge operation enables GraphRAG to adeptly manage duplicate geomorphological entities and to consolidate the graphs associated with multiple text chunks into a novel graph, subsequently summarizing the descriptions of the nodes and relationships within this new graph. This streamlining aids in subsequent queries by enabling the matching of geomorphological entity and relationship information within the knowledge base, relying exclusively on the summarized descriptions of entities and relationships.

2.2 Geomorphology Knowledge Graph Community

The hierarchical Leiden algorithm is employed to construct the hierarchical structure of entity communities, delivering multi-scale community detection outcomes that augment the comprehension of the geomorphology

knowledge graph. During query execution, by designating the community level, one can differentiate the search results accordingly. For instance, in response to the query "What are the types of sedimentary landforms?", if the loess landforms community is specified, the results would encompass loess valley landforms, loess interfluvial landforms, and loess subsurface erosion landforms. Conversely, if the wind-deposited landforms community is specified, the results would extend beyond loess landform types to include other varieties such as sand dunes. The algorithm operates by recursively clustering communities, refining the modularity Q at each hierarchical level until a predefined community size threshold is attained. Modularity Q is quantified as follows:

$$Q = \frac{1}{2m} \sum_{ij} (A_{ij} - \frac{k_i k_j}{2m}) \delta(c_i, c_j)$$

A_{ij} denotes an element of the adjacency matrix, signifying the weight of the edge connecting node i and node j . The terms k_i and k_j represent the degrees of node i and node j , respectively, while m signifies the aggregate number of edges within the graph. The function $\delta(c_i, c_j)$ serves as an indicator, taking the value of 1 if nodes i and node j are members of the same community, and 0 otherwise.

Community summaries utilize embedding vectors to precisely retrieve geomorphological entities, thereby identifying pivotal nodes and subgraph configurations within the knowledge graph. By employing large language models, summaries are crafted for each graph community, with predefined prompts facilitating a hierarchical, top-down summarization of the communities. This methodology aids in comprehending the overarching thematic architecture and semantic nuances of the dataset, allowing for the citation of crucial entities, relationships, and propositions within the subgraphs of the graph communities. The Node2Vec algorithm is applied to transform the generated community reports, their summaries, and the titles into vectorized forms, thereby producing vector representations of the graph and establishing an auxiliary vector space. This enhancement enables the expedient querying of associated concepts during subsequent retrieval operations.

2.3 Retrieval Augmentation Based on Geomorphology Knowledge Graph

Within the GraphRAG framework, retrieval augmentation assumes a pivotal role. This technique synergizes graph structures with external knowledge repositories to markedly bolster the model's efficacy and precision in tackling intricate tasks. Retrieval augmentation achieves this by assimilating pertinent data from external knowledge sources that are associated with nodes, thereby facilitating the grounding of large language models within the knowledge graph. It accomplishes this by indexing the original text to furnish answers that are not only more dependable but also accompanied by their provenance, thus enabling the validation and trustworthiness of the outputs produced by large language models. By harnessing knowledge graphs that are the product of large language models, GraphRAG substantially refines the "retrieval" component of RAG, populating the context window with content of heightened relevance, which in turn leads to superior answers and the identification of evidentiary sources.

At the heart of retrieval augmentation is the utilization of graph neural network models to capture the local context of nodes, amalgamated with data from external knowledge repositories to produce enriched entity

representations. This procedure is depicted in Fig. 4. Upon a user posing a question like "What is the cause of landforms?", the system discerns a cluster of entities from the geomorphology knowledge graph that bear semantic relevance to the query, such as "landform causes." These entities act as conduits into the geomorphology knowledge graph, whereupon the target entities along with their proximate nodes are retrieved to glean additional pertinent details. Entities are then correlated with corresponding text segments, encompassing relevant excerpts from source documents, community reports, attributes of nodes, and inter-node relationships. The assembled text is subsequently prioritized, with various informational elements being assigned weights to underscore and seamlessly amalgamate critical data. This synthesized textual information is then embedded within a prompt framework and fed into the large language model, which is thereby empowered to craft responses that are both more nuanced and precise.

3 Experiment and result

3.1 Experiment design

This study leverages the GraphRAG framework to embed geomorphological knowledge within large language models, thereby augmenting the profundity and scope of geomorphological inquiry. To assess the proficiency of large language models enhanced by geomorphology knowledge graph retrieval in the comprehension and identification of geomorphological issues, the investigation adopts karst landform knowledge as an exemplar and employs a dataset comprising geomorphology-centric questions from the national postgraduate entrance examinations conducted over the preceding three years. The dataset is predominantly composed of inquiries that revolve around geomorphological principles and subject matter, from which pertinent geomorphological knowledge questions are extracted for evaluative scrutiny. The research establishes linkages between the knowledge repository and the large language models via both RAG and GraphRAG frameworks, facilitating a comparative analysis of the resultant output quality.

Taking into account factors such as efficacy and the exigencies of the task at hand, the study selects ChatGLM3-6B as the foundational model, utilizing a compendium of 2000 scholarly articles and 22 authoritative texts on geomorphology as the primary corpus, which are subsequently vectorized to construct an external vector database.

3.2 Evaluation Methods

(1) Subjective Evaluation

Following a comparative examination of various retrieval methodologies applied to identical queries, this study employs a random sampling technique to select 50 geomorphological knowledge essay questions from the question bank for each iteration. Utilizing reference answers as a benchmark, three testing cycles are executed to juxtapose the outcomes produced by distinct retrieval approaches in question answering. Responses are categorized into three groups based on their alignment with the reference answers: positive enhancement, basic similarity, and negative enhancement. Positive enhancement responses are distinguished by their precise coverage of the question's focal points and their superior depth, breadth, or logical organization compared to the reference answers, thereby furnishing users with more comprehensive replies. Basic similarity responses are characterized by their substantial congruence with the reference answers in essential

content and framework, differing only in slight expressive nuances, thus fulfilling fundamental question-answering requirements without notable enhancements. Negative enhancement responses are identified by their discernible decline in the accuracy, thoroughness, or logical flow of information relative to the reference answers, possibly harboring deceptive content or omitting pivotal elements.

(2)Objective Evaluation

To assess the professional efficacy of large language models employing the GraphRAG architecture in the domain of geomorphological knowledge question answering, this study adopts four metrics from the RAGAs framework: Faithfulness, Answer Relevance, Context Precision, and Context Recall. These metrics serve as evaluative benchmarks to gauge the extent of knowledge coverage and the pertinence of the knowledge within the generated responses.

The Faithfulness metric evaluates the factual alignment of the generated answers with the provided context. It is computed by analyzing the extent to which the claims in the answers are supported by the retrieved context. Specifically, the metric quantifies the proportion of claims in the response that can be substantiated by the context. If all claims in the answer can be logically inferred from the given context, the answer is considered faithful. To calculate this metric, the following steps are taken: first, a set of claims is extracted from the generated answer; then, each claim is rigorously compared against the context to verify its derivability. The Faithfulness score is determined using the formula:

$$Faithfulness\ score = \frac{|Number\ of\ claims\ in\ the\ response\ supported\ by\ the\ retrieved\ context|}{|Total\ number\ of\ claims\ in\ the\ response|}$$

The Answer Relevance metric quantifies the relevance of the generated answer to the posed question. This metric is derived by inversely deducing potential variant questions from the answer and then calculating the cosine similarity between these inferred questions and the original question. This process evaluates whether all the information contained within the original answer is pertinent to the question. Should the original answer be lacking in completeness or include extraneous details not pertinent to the question, the inversely deduced questions may omit critical information or include superfluous details, thereby diminishing the Answer Relevance score. The formula for calculating this metric is presented below:

$$Answer\ Relevancy = \frac{1}{n} \sum_{i=1}^N \cos(E_{gi}, E_0)$$

E_{gi} represents the embedding vector of the variant question that is inversely deduced from the answer, while E_0 denotes the embedding vector corresponding to the original question.

Context Precision serves to assess the degree of relevance between the retrieved context and the question, encompassing both the pertinence of the context to the question and the order in which the context is presented. The formula for calculating Context Precision is as follows:

$$Precision_k = \frac{TP}{TP + FP}$$

$$\text{Context Precision}_k = \frac{\sum_{k=1}^K (\text{Precision}_k \times v_k)}{R_k}$$

In this context, Precision_k denotes the ratio of segments pertinent to the question within the top k segments. TP quantifies the instances where the model accurately employs the retrieved relevant context information, whereas FP tallies the instances where the model erroneously employs context information that is either irrelevant or misleading. The term v_k signifies the degree of relevance or significance attributed to the k -th retrieval outcome, and R_k enumerates the aggregate count of relevant segments within the top K retrieval results.

Context Recall measures how many of the relevant documents (or pieces of information) were successfully retrieved. It focuses on not missing important results. Higher recall means fewer relevant documents were left out. In short, recall is about not missing anything important.

$$\text{Context recall} = \frac{GT'}{GT}$$

GT represents the aggregate count of assertions within the ground truth, while GT' signifies the subset of these assertions that are accurately ascribable to the context that has been retrieved.

3.3 Results and analysis

The experiment evaluates the efficacy of large language models integrated with RAG against those integrated with GraphRAG in the realm of geomorphological knowledge question answering, utilizing a consistent dataset and evaluation criteria. Table 1 delineates the text retrieved by disparate retrieval methodologies for identical queries. In the retrieval of text for the same query using the two distinct methods, RAG's similarity search for "tower karst" yielded only the genesis and fundamental definition of karst, as the specific term was absent from the dataset, culminating in an unresponsive answer. Conversely, GraphRAG's retrieval capitalized on the knowledge graph's stored equivalence between "tower karst" and "peak forest," thereby broadening the search to encompass the genesis of peak forest in alignment with the original query. This approach retrieved pertinent geomorphological texts on peak forest formation, culminating in an answer that elucidated the nexus between tower karst and peak forest, along with the formative processes of tower karst. The knowledge graph's repository of explicit geomorphological relationships facilitates the augmentation and refinement of queries during retrieval, thereby empowering the large language model to deliver more adept responses to geomorphological inquiries.

Table 1
Recall of Texts by Different Retrieval Methods

Retrieval Method	Question: What are the causes of the formation of tower karst?
	Recall Text
RAG	(1)Structure of karst landforms: Section 1 Definition of karst; Section 2 Conditions and factors influencing karst landform development; Section 3 Fundamental principles of karst landform development and considerations of the karst base level... (2)The origin of "karst": The term "karst" (Karst) originally denoted a geographical region, specifically the Karst Plateau in the northwestern Dinaric Alps, straddling the border between Yugoslavia and Italy. This limestone-rich area exhibits a variety of distinctive landforms resulting from dissolution and erosion processes, intersected by several canyons. (3)The fundamental definition of karst: As scientific understanding has evolved, interpretations of "karst" have varied among scholars. Numerous definitions have been proposed, which can be broadly categorized into three main perspectives. The first perspective regards karst as a "process," specifically the geological processes resulting from the interaction of water with soluble rocks...
GraphRAG	(1)The structure of karst landforms: Section 1 Definition of karst; Section 2 Conditions and factors influencing karst landform development; Section 3 Fundamental principles of karst landform development and considerations of the karst base level... (2)The genesis of peak clusters: The formation and morphology of peak clusters are influenced by tectonic structures. Peak clusters are typically aligned with the strike of rock layers. At the anticlinal axis, peak clusters form more readily and develop more completely. In this context, due to the approximately horizontal strata, peak clusters are generally taller, vertical, and symmetrical. On the anticlinal limbs, peak clusters often assume the form of cuestas, with reduced heights, resulting in less prominent isolated stone peaks (with dip angles exceeding 25°) or forming serrated ridges that extend along the strike of the strata. (3)The influence of climate on peak clusters: Climate plays a crucial role in the development of peak clusters. In tropical and monsoon regions, such as the tropical monsoon rainforest and equatorial rainforest zones characterized by perpetual heat and rainfall, dissolution processes are significantly more intense than in temperate or Mediterranean climates. Consequently, peak cluster landforms are extensively developed in warm and humid climatic regions...

The two retrieval methods exhibit notable disparities in their responses to identical queries, as delineated in Table 2. Addressing the query "Elucidate the factors influencing landform formation across diverse climatic conditions," RAG-based responses are confined to the pertinent textual data within the dataset, delineating the attributes of the "glacial climate landform zone," "temperate humid climate landform zone," "arid climate landform zone," and "humid tropical climate landform zone." These responses are devoid of synthesizing commentary and fail to encapsulate the profound interplay between climatic conditions and landform genesis. Conversely, GraphRAG, by virtue of the geomorphology knowledge graph's modular architecture, attains a holistic comprehension of the dataset. The resultant responses encompass not only the distinctive features of landforms within various climatic zones but also furnish a synthesized discourse on the nexus between climatic conditions and landform formation, thereby rendering the answers more cohesive and logically structured. This underscores that, in contrast to RAG-derived responses, GraphRAG's methodology, anchored in the geomorphology knowledge graph's structural attributes, excels in retrieving text of greater veracity, thereby enhancing the trustworthiness of the generated content. Furthermore, through the strategic formulation of prompts, it is capable of eliciting responses that are both more expert and encompassing.

Table 2
The Effectiveness of Different Retrieval-Augmented Methods in Answering Questions

Retrieval Method	Question: What are the factors contributing to geomorphological formation under different climatic conditions?
	Answer
RAG	The formation of landforms under different climatic conditions is primarily governed by the physical and chemical processes acting on surface materials, which are influenced by climate. Below are several typical climatic zones and their associated landform characteristics: Glacial Climate Landform Zone: ((1) Glacial climate zone: Found in high-latitude polar regions and areas above the snowline in high mountains, where average annual temperatures are below 0°C, and the landscape is perpetually covered by ice and snow. Glacial processes dominate, creating landforms such as cirques and U-shaped valleys. (2) Periglacial climate zone: Located in ice-free polar and subpolar regions, as well as high-altitude areas below the snowline but above the treeline. Here, average annual temperatures hover near 0°C, and seasonal freeze-thaw cycles drive the development of permafrost and periglacial landforms. Temperate and Humid Climate Landform Zone: (1) Predominantly found in mid-latitude regions, with average annual temperatures around 10°C and annual precipitation of approximately 800 mm. Fluvial processes are highly active in this zone, shaping landforms such as river terraces and canyons. (2) In the eastern continental regions, monsoonal influences concentrate precipitation during summer, intensifying erosion and deposition processes and producing distinctive landform features. (3) In the western regions, maritime climates ensure more evenly distributed precipitation, resulting in milder erosion and deposition. This allows ancient landforms to remain well-preserved. Arid Climate Landform Zone: (1) Situated in subtropical high-pressure zones and the interiors of temperate continents, these regions experience extreme aridity, with annual precipitation typically below 250 mm and evaporation rates far exceeding precipitation. (2) Sparse vegetation and exposed ground surfaces lead to intense physical weathering, with wind erosion becoming the dominant geomorphic force. This results in aeolian landforms such as deserts and gravel plains (gobi). (3) The semi-arid transitional zone between arid and humid regions receives about 400 mm of annual precipitation and is characterized by loess deposits, which give rise to unique loess landforms. Hot and Humid Climate Landform Zone: (1) Located near the equator and in low-latitude regions, these areas experience average annual precipitation exceeding 1000 mm and average temperatures above 18°C in the coldest month, with no true winter season. (2) High temperatures and abundant rainfall drive intense biochemical weathering, producing thick lateritic weathering profiles.

Retrieval Method	Question: What are the factors contributing to geomorphological formation under different climatic conditions?
	Answer
GraphRAG	The formation of landforms is intricately linked to climate, as climate plays a pivotal role in this complex process. Climatic conditions, particularly temperature and precipitation, directly determine the nature and intensity of geomorphic processes while indirectly influencing landform development through vegetation cover and the hydrological cycle. Landform characteristics in hot and humid climates: In hot and humid climatic zones, such as equatorial regions, high temperatures and abundant rainfall enhance chemical and biological weathering, accelerating rock breakdown. In these environments, fluvial erosion is particularly pronounced, leading to the formation of deep river valleys and extensive karst landscapes, exemplified by the karst topography in Guilin, China. Landforms in arid and semi-arid regions: In arid and semi-arid zones, limited precipitation makes wind the dominant geomorphic agent, creating aeolian landforms such as desert sand dunes. In these areas, where evaporation rates far exceed precipitation, the development of weathering crusts and wind-driven erosion results in distinctive landform features. Landforms in cold climates: In cold climates, glacial processes dominate landform shaping. Glacial erosion and deposition produce features such as U-shaped valleys, moraines, and glacial lakes in high-altitude and high-latitude regions. Additionally, periglacial processes, including freeze-thaw cycles, significantly influence landform development. Comprehensive factors in landform formation: Landform formation is not driven by climate alone but by the interplay of internal forces (e.g., tectonic activity) and external forces (e.g., erosion, sedimentation, weathering) under varying climatic conditions. Over time, landforms evolve through distinct developmental stages, each influenced by the preceding stage. Human activities, such as urbanization and land modification, have also emerged as significant contributors to contemporary landform changes. In summary, climate critically shapes the formation and distribution of landforms by governing the types and intensity of geomorphic processes. From the karst landscapes of hot and humid regions to the glacial-sculpted features of cold zones and the aeolian landforms of arid areas, each climatic regime creates unique landform characteristics. This diversity underscores the complexity and richness of Earth's natural systems.

3.4 Results Evaluation

(1)Subjective Evaluation

The comparative assessment of various retrieval augmentation methodologies is depicted in Fig. 5. Across three iterations of question-answering trials, the GraphRAG approach consistently exhibited a superior positive enhancement rate relative to the conventional RAG method, underscoring GraphRAG's pronounced efficacy in elevating the caliber of replies to inquiries pertaining to geomorphological knowledge. It is noteworthy that in the generation of non-negative enhanced replies, both RAG and GraphRAG, augmented with the geomorphology knowledge graph, performed comparably. This outcome substantiates that the retrieval technique integrating the geomorphology knowledge graph adeptly preserves the fundamental response

quality sans inducing supplementary adverse enhancement impacts. Subjective appraisals over several question-answering experiment cycles facilitated a multidimensional analysis, encompassing the precision of the retrieved text, the diversity of the produced answers, and the positive enhancement rate. The GraphRAG methodology distinctly excelled in the realm of knowledge abundance. The empirical findings reveal that, in contrast to the traditional RAG approach, the retrieval augmentation tactic fused with the geomorphology knowledge graph is more adept at infusing specialized domain knowledge into large language models, consequently enhancing the substance of the generated replies.

(2)Objective Evaluation

The efficacy of various retrieval augmentation techniques is illustrated in Fig. 6. GraphRAG's average scores across four key metrics predominantly fall within the medium to high spectrum, suggesting that the assimilation of geomorphological knowledge via knowledge graph integration yields responses with enhanced trustworthiness and precision, which are more pertinent to the queries and embody the specialized nuances of geomorphology. This substantiates the utility of engaging with the geomorphology knowledge graph in tackling geomorphological inquiries. Nonetheless, a granular analysis of specific metrics reveals that GraphRAG marginally trails RAG in context precision and context recall. This discrepancy is primarily due to GraphRAG's retrieval framework, which, by virtue of its graph-centric approach, yields a voluminous set of potentially pertinent graph data. Such data may be deemed unrelated to the queries during assessment, consequently diminishing the associated metric scores. Conversely, RAG's retrieval methodology is more straightforward, yielding content that is more aligned with the queries, hence its superior evaluation performance. Despite GraphRAG's prowess in discerning intricate relationships, the expansiveness and intricacy of its retrieval outcomes impinge upon the evaluation metric performance.

In summary, by juxtaposing a range of evaluation metrics across diverse retrieval methodologies, it is evident that the GraphRAG-based approach to geomorphological knowledge question-answering presented in this study excels in terms of response credibility, relevance, and accuracy, and exhibits a higher degree of knowledge richness and overall performance. Consequently, the proposed methodology is adept at augmenting the efficacy of large language model retrieval question-answering systems, particularly in the realm of geomorphological expertise.

4 Discussion and Conclusion

GraphRAG introduces an innovative methodology for the acquisition, storage, management, and presentation of geomorphological knowledge, facilitating an intelligent question-answering framework that seamlessly integrates knowledge graphs with large language models. This study centers on geomorphological knowledge as its research focus, delving into the collaborative dynamics between knowledge graphs and large language models, and introduces a GraphRAG-driven approach for geomorphological knowledge question answering. Employing karst landform knowledge as an illustrative case, the experimental findings reveal that the proposed method surpasses traditional RAG techniques in both accuracy and comprehensiveness of contextual knowledge retrieval, especially in tackling intricate issues characterized by elevated levels of abstraction and generalization within geomorphology. Assessments grounded in the profundity and expertise of geomorphological knowledge question answering corroborate the efficacy and utility of the proposed method.

While the existing geomorphology knowledge graph is capable of facilitating a range of question-answering tasks, there remains significant potential for augmenting its breadth and depth. Prospective advancements may involve the incorporation of additional high-caliber geomorphological data sources, including but not limited to geological survey data and remote sensing data, to broaden the knowledge graph's scope. Furthermore, the adoption of state-of-the-art graph embedding methodologies could refine the knowledge graph's representational efficacy, thereby bolstering the question-answering system's proficiency and equipping it with more prescient intelligent tools to address intricate geomorphological scientific challenges.

Declarations

Competing interests

The authors declare no competing interests.

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Author Contribution

X. Zhang and J. Du wrote the main manuscript text. J. Du and B. Guo developed the code and conducted the experiments. B. Guo and M. Xiang prepared all figures and tables. All authors reviewed the manuscript.

Data Availability

The experimental code supporting the paper has been included in the file Code.rar.

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Figures

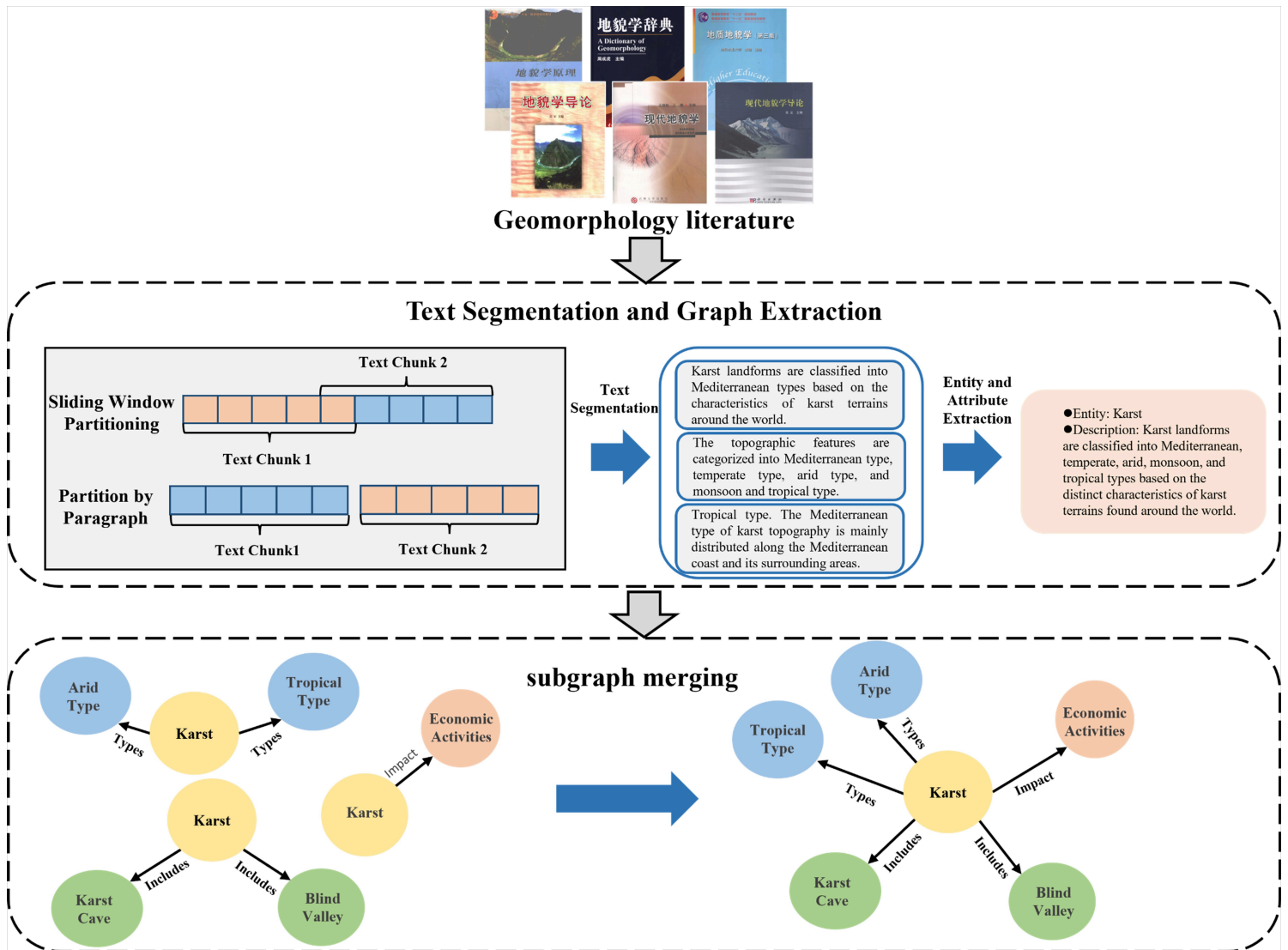


Figure 2

Geomorphology Knowledge Graph Construction Process

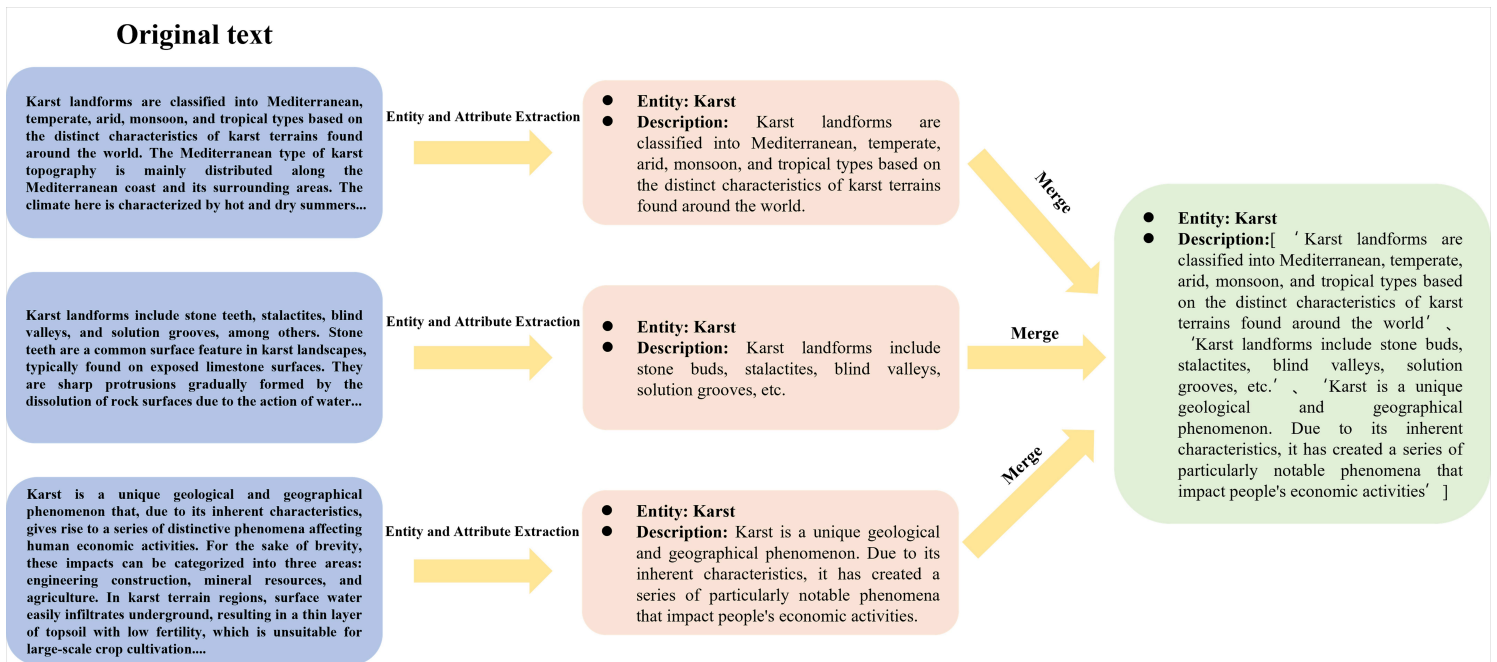


Figure 3

GraphRAG Geomorphology Knowledge Graph Merging Example

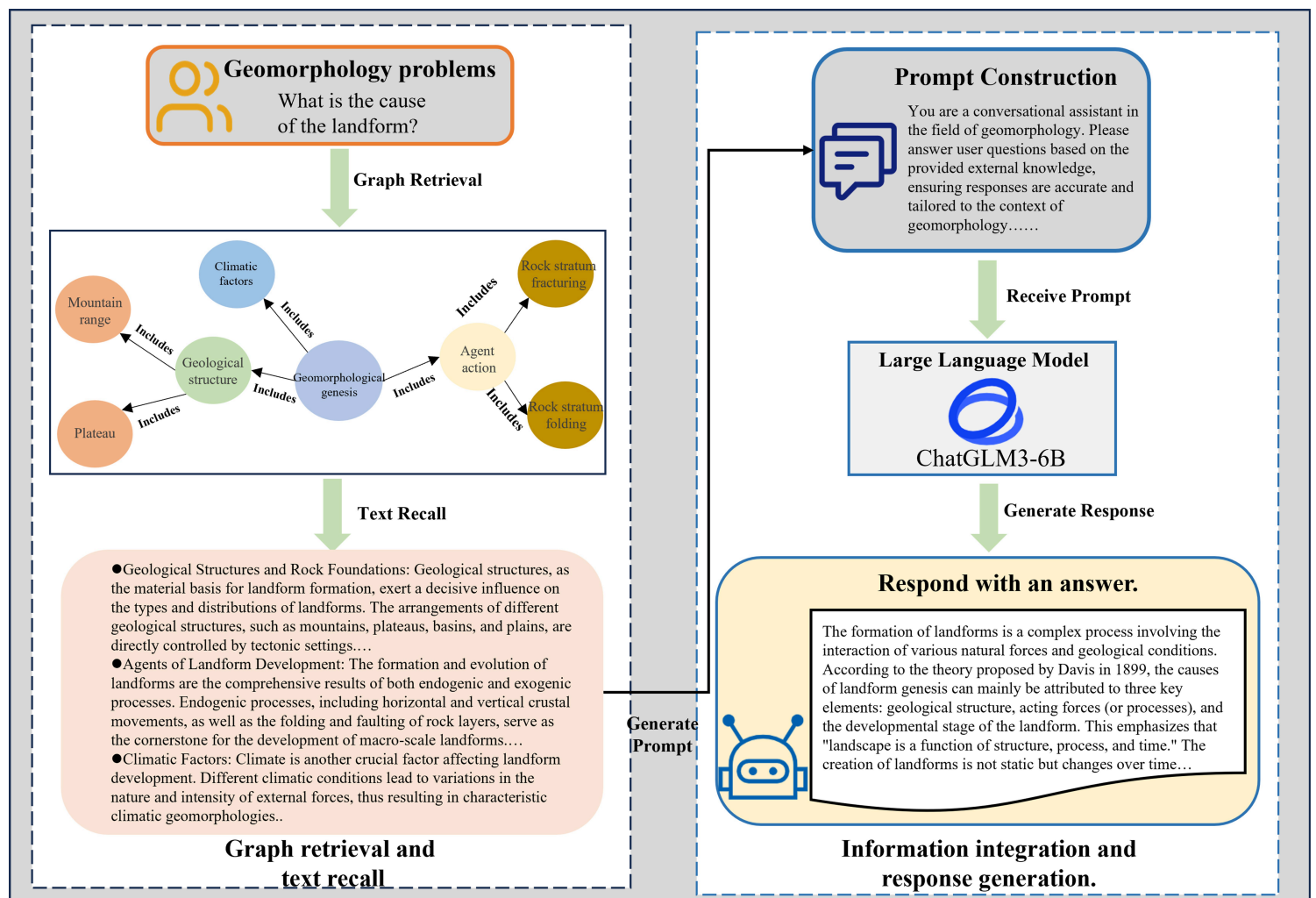


Figure 4

Retrieval Augmentation Process

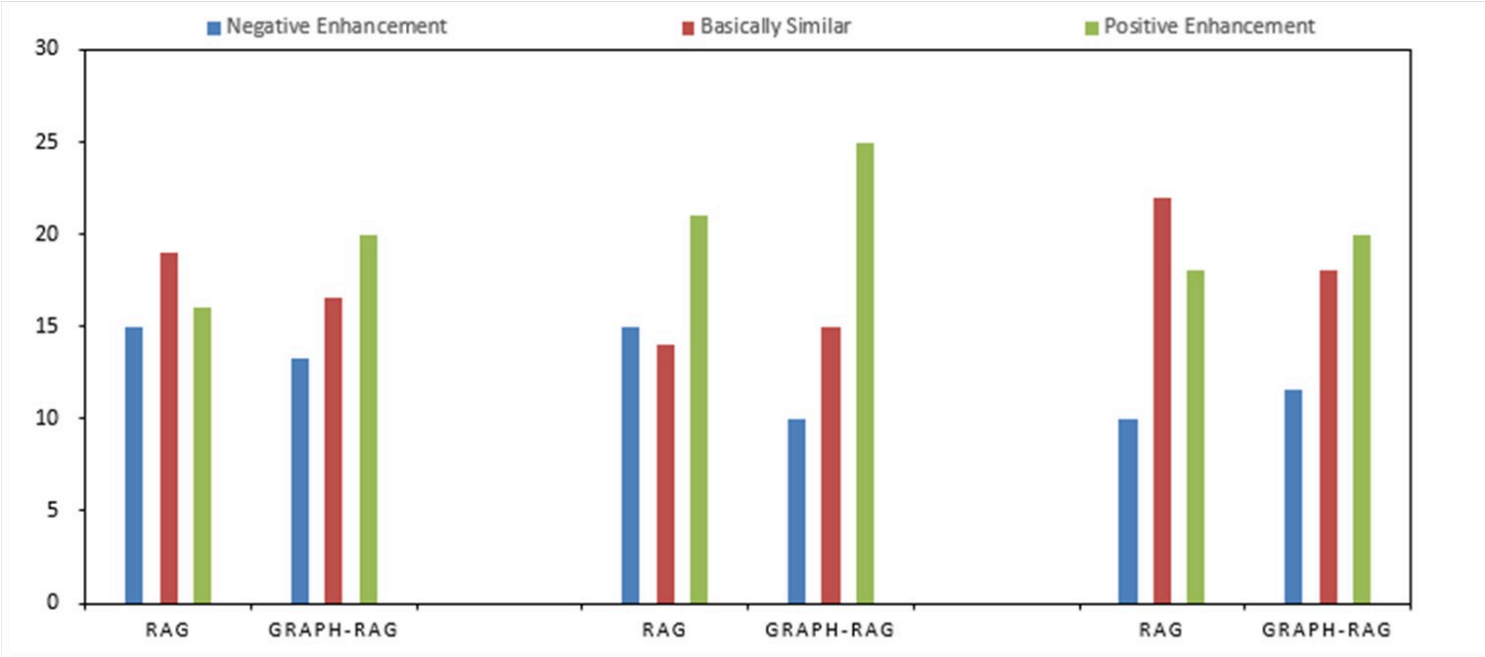


Figure 5

Evaluation Experiment of Different Retrieval Augmentation Methods

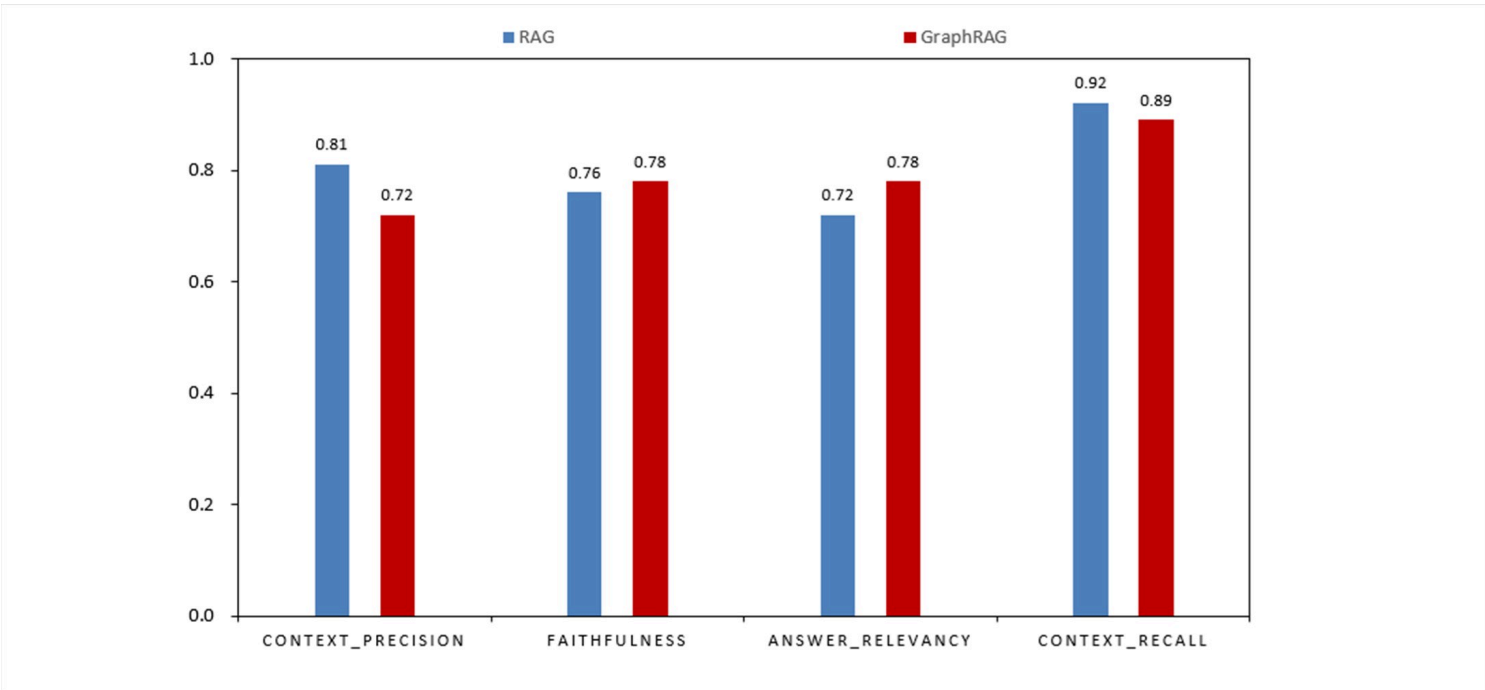


Figure 6

Evaluation Metrics for Different Retrieval Augmentation Methods